My journey into AI and machine learning started with moments that made me think, "this should work better," across different domains.

The inventory management system I built for our robotics lab began as a simple Flask and MySQL tracking tool. Frequent shortages of critical components motivated me to integrate a demand forecasting model that predicts restocking needs based on historical usage patterns. Today, that system supports more than ten student projects in our lab. This experience taught me that machine learning becomes much more meaningful when its reliability directly impacts others' work.

My most significant project has been an interpretable machine learning framework for distinguishing epilepsy from psychogenic non-epileptic seizures. I used structured clinical data from 271 EMU admissions to evaluate Logistic Regression, Random Forest, XGBoost, TabNet, and FT-Transformer models. I found that SMOTE, despite being a common solution for class imbalance, reduced ROC-AUC by 0.08. In contrast, optimizing the threshold on the precision-recall curve preserved epilepsy recall without adding synthetic bias. An early version of the model achieved a suspiciously perfect AUC of 1.0. I investigated it through five explainability methods: SHAP, LIME, Anchor, QLattice, and TabNet. This revealed target-feature leakage that conventional validation had failed to detect. Fixing that issue and seeing all five methods converge on EEG findings as the main predictor taught me that careful experimental design and interpretability are just as important as model performance.

Beyond healthcare, I developed a traffic demand prediction pipeline using geohash-based spatial features, cyclical time encodings, and ExtraTreesRegressor, achieving R^2 scores above 0.90. I also designed an efficient drone-delivery routing algorithm by replacing brute-force permutation searches with spatial grid indexing and a min-heap scheduler, leading to significant computational speedups.

Throughout these projects, I have faced the same limitation repeatedly: while I can build effective models and explain their behavior after the fact, I want the theoretical background to understand why a method should generalize beforehand. Areas like statistical learning theory, optimization, and representation learning are where working with active researchers can provide insights that experimentation alone cannot. The leakage issue in my epilepsy research illustrates this while interpretability tools revealed the problem, a deeper theoretical understanding could have led to a stronger experimental design from the start.

MLSS offers the environment I am looking for: immersion in the mathematical foundations and research frontiers of machine learning alongside researchers and peers who challenge assumptions and foster deeper thinking. My projects cover clinical AI, inventory forecasting, spatial optimization, and time-series prediction, reflecting a curiosity that naturally crosses disciplinary boundaries. I hope to bring that breadth of perspective to MLSS while gaining the theoretical depth needed to pursue impactful and rigorous machine learning research.